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Paleomagnetic reconstruction of the Cenozoic evolution of the Eastern Mediterranean

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Abstract

A total of 113 paleomagnetic sites were sampled along an Anatolian S–N transect from the Arabian platform, the Hatay region, the Eastern Taurides, the Kirsehir block, the Sivas basin and the Eastern Pontides. Reliable characteristic remanent paleomagnetic directions were retrieved from 37 of these sites, spanning in time from Paleocene to Miocene. In a general way, declinations are westerly deviated and inclinations are shallower than the geocentered dipole value at the present latitudes. When combined with previously published results, these data indicate that a large-scale counterclockwise rotation of Anatolia of some 25° has occurred since the Miocene. Assuming that the pole of rotation of Anatolia with respect to Europe has remained constant in time at the location given by MacClusky et al. [J. Geophys. Res. 105 (2000) 5695] on the basis of the geodetic data, this rotation implies that a large westward displacement (500 km at the average latitude of 40°) has taken place. Assuming that the rotation was initiated by the Arabia/Europe collision about 12 Ma ago, this corresponds to an average displacement of about 40 mm/year.

Together with previous results from the western part of the Aegean arc, these results indicate that the main trends of the Cenozoic evolution of the Eastern Mediterranean appear to consist of two post-early Miocene rotations of opposite senses: a clockwise rotation of the western part of the Aegean [Tectonophysics 146 (1988) 183] around a pole situated in northern Albania, and a counterclockwise rotation around the pole given by McClusky et al. [J. Geophys. Res. 105 (2000) 5695]. Comparison with GPS data suggest that both rotations are still active today.
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1. Introduction

The Mediterranean Alpine chain, which results from the convergent movement of the two major African and European plates, is one of the most tectonically active and mobile mountain belts. Its formation started during Mesozoic time by the closure of the Tethyan ocean and by progressive accretion of blocks to the Eurasian plate. It is still active today as evidenced by the present seismic activity.

Among the tectonic, stratigraphic and geophysical data used to develop models describing the evolution

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of this belt, paleomagnetic data are crucial because they allow a quantitative estimation of the role of both latitudinal drift and rotations around vertical axes in the building of the orogens. Despite the interest of such studies, the eastern Mediterranean and in particular Anatolia have been poorly studied from a paleomagnetic point of view before the 1980s, especially when compared to the large dataset available from western and central Mediterranean. Except for a study of Eocene rocks in northern Turkey by Van der Voo (1968), paleomagnetic studies were undertaken mostly in Permo-Triassic and Mesozoic units. The interpretation of the rotations observed in these old formations in terms of recent movements is, however, difficult because the recent tectonic activity of these areas may also have induced movements of large amplitude. During the 1980s, paleomagnetic results obtained from Eocene to Miocene rocks from southern and western Turkey were interpreted in terms of evolution of the Aegean domain (Kissel and Poisson, 1986, 1987; Kissel et al., 1993). Saribudak (1989) also produced paleomagnetic data from the western and Central Pontides. More recently, several authors have investigated areas in the vicinity of the major faults in order to determine the amount of tectonic deformation across the fault zones (Platzman et al., 1994; Tatar et al., 1995, 2000; Piper et al., 1996; Gürsoy et al., 1997). The widespread Pliocene to Quaternary volcanic formations from Central Anatolia were also studied in order to constrain the timing of the rotation (Tatar et al., 1996; Gürsoy et al., 1998; Platzman et al., 1998).

In this paper, we report on the paleomagnetic results obtained from Paleocene to Miocene units distributed along a N–S transect in eastern and central Turkey. These paleomagnetic data, combined with data previously published by other authors, allow to propose a model for the geodynamic evolution of eastern Mediterranean which is compared with the results of recent geodetic data.

2. Geodynamical and geologic setting

The Eastern Mediterranean Alpine belt results from the convergent movement of the Laurasian and the Arabia/Africa plates. During the Mesozoic time, the Arabia/Africa plate moved northwestward and progressively attained its present longitudinal position with respect to Eurasia in the Late Cretaceous (Dercourt et al., 1993; Görür, 1998). During this period, rifting and drift of continental fragments detached from the Gondwanian margin created new basins in several steps (Late Permian, Triassic, Early Cretaceous). At the end of the Cretaceous, Eurasia and Africa initiated a north–south convergent movement. Along the Turkish transect, the oceanic basins progressively declined first by subduction and subsequently by obduction onto the neighboring continents. A review of the models can be found in Robertson and Dixon (1984) and Robertson et al. (1996). Among these models, the plate tectonic reconstruction of Sengör and Yilmaz (1981) has been a main reference for the tectonic evolution of Turkey and the Eastern Mediterranean. The northern Neotethys closed definitely between the Campanian and the Late Paleocene (Okay and Tüysüz, 1999), while south of the Anatolian block, the Pamphylian basin also initiated its completion by the southward obduction onto the Cyprus basement and the Arabian plate, but its complete closure occurred as late as Eocene (see, e.g. Poisson, 2003; Robertson, 2000). All the tectonic units derived during this period were unconformably overlain by Late Paleocene, Early Eocene or the Lutetian shallow marine deposits.

The main structural units underlined by the major discontinuities corresponding to the Neotethyan sutures, from North to South, are discussed below.

· The Pontic belt represents the southern margin of the Eurasian plate which collided with the Anatolian microcontinent when the North Neotethyan ocean sutured in Late Cretaceous–Paleocene times (Yilmaz et al., 1997; Okay and Tüysüz, 1999).

· The Anatolian block separated from the Pontides by the Izmir–Ankara–Erzincan–Erzurum suture zone is composed mainly of ophiolitic nappes and ophiolitic melanges of Late Cretaceous age (Sengör and Yilmaz, 1981). Along the Central Anatolian transect considered here, the Anatolian block consists of two main continental fragments: the Kirsehir Massif and the Taurus platform, separated by the Sivas Tertiary basin. The metamorphic sequence of the Kirsehir massif has been intruded by various plutonic rocks of middle Cretaceous to Paleocene age (Göncüoglu and Türeli, 1994; Boztug, 1998). The main Eocene and Miocene deposits are dominantly continental with the exception of the Nummulitic sandstone/lime-

stones. The rest of the massif is largely covered by Quaternary lavas and ignimbrites (Innocenti et al., 1975; Pasquare et al., 1988; Göncüoglu and Toprak, 1992; Platzman et al., 1998). The Taurus carbonate platforms are structurally complex. They include at least two allochthonous nappes above an autochthonous series represented by the Sariz–Gurun unit. The Sivas Basin is one of the central Anatolian basins formed over part of the Pontic belt, the ophiolites, the Kirsehir Massif and the Taurides after the closure of the North Neotethys ocean (Sengör and Yilmaz, 1981). Its infill derives from the erosion of all these neighboring areas from Paleocene to Miocene.

· The northern part of the Arabian plate crops out in SE Turkey. The general organization of this area during Mesozoic and Tertiary times is that of a carbonate ramp transitional from a continental area to the south (the Arabic craton) to a deep basin to the north.

The Late Miocene and the period from the Pliocene to the present are usually associated in a single "neotectonic period" during which a major dislocation of entire Anatolia occurred along two prominent active structural lines: the dextral North Anatolian and the sinistral East Anatolian strike-slip fault zones (NAFZ and EAF). This fragmentation is coeval with an opening pulse of the Red Sea, which initiated about 12 Ma ago. Between these faults, the Central Anatolian Block moves westward. These fault zones are several hundred kilometers long, and their many strands cut through various units. The initiation of the motion along the faults and the evaluation of the total lateral offsets remain an open question and various estimations are still in discussion due to the lack of constraining data except in some restricted areas (Sengör, 1979; Sengör and Canitez, 1982; Saroglu, 1988; Rotstein, 1984).

3. Paleomagnetic sampling

We have sampled 113 paleomagnetic sites (1300 oriented cores) in three successive field trips in 1986, 1987 and 1988 along a N–S transect from the Pontides to the Arabian plate in different tectonic units (Fig. 1).

Eighteen sites were sampled in Eocene and Miocene volcanics and limestones in the Gumushane area in the Eastern Pontides (Fig. 2). Sediments interbedded with lava flows were used to determine the bedding plane.

In the Kirsehir massif, five sites were sampled in the instrusive rocks near the town of Kaman. Three

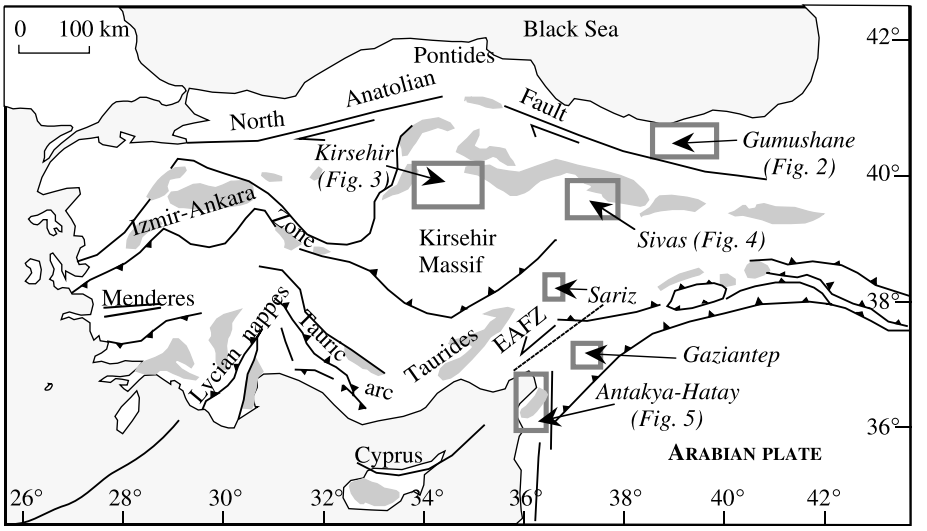


Fig. 1. Schematic map of Turkey showing the location of the sampled regions. The gray areas correspond to the main ophiolitic nappes. EAFZ: East Anatolian Fault Zone.

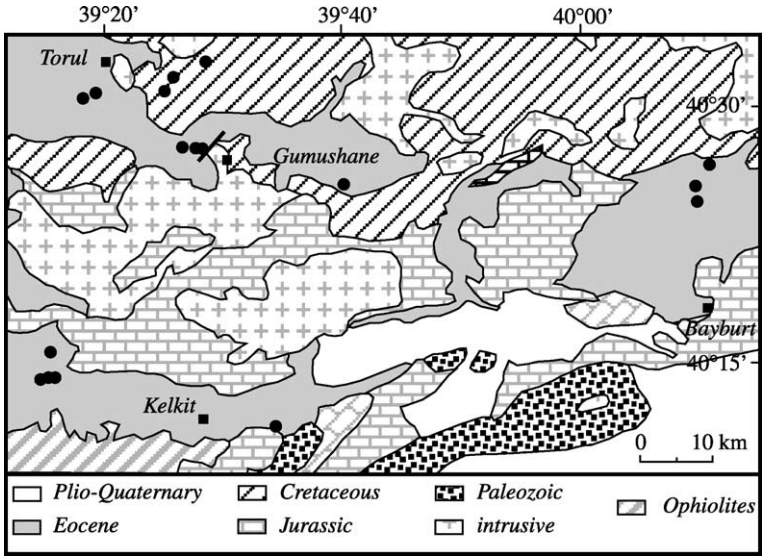


Fig. 2. Schematic map of the Gumushane area in the Eastern Pontides showing the location of the sampled sites.

sites in the Cretaceous intrusive outcrops and nine Eocene lava flows and interbedded sediments were sampled close to the town of Yozgat (Fig. 3).

We sampled 10 Cretaceous to Eocene sites along a section on the flank of an anticline close to Sariz (Fig. 1). This is part of the broad antiform striking NE–SW which constitutes the autochthonous pile of the Taurus carbonate platform.

In the Sivas Basin, 17 sites were sampled in the Sarkisla–Zara region, in Paleocene volcano-sedimentary formations, limestones and lavas (Fig. 4).

Finally, we have sampled the northern part of the Arabian plate near Gaziantep and Antakya. In the Gaziantep area, we sampled six sites in upper Cretaceous limestones, five sites in Paleocene blue-grey marls and finally two sites from Miocene volcanics

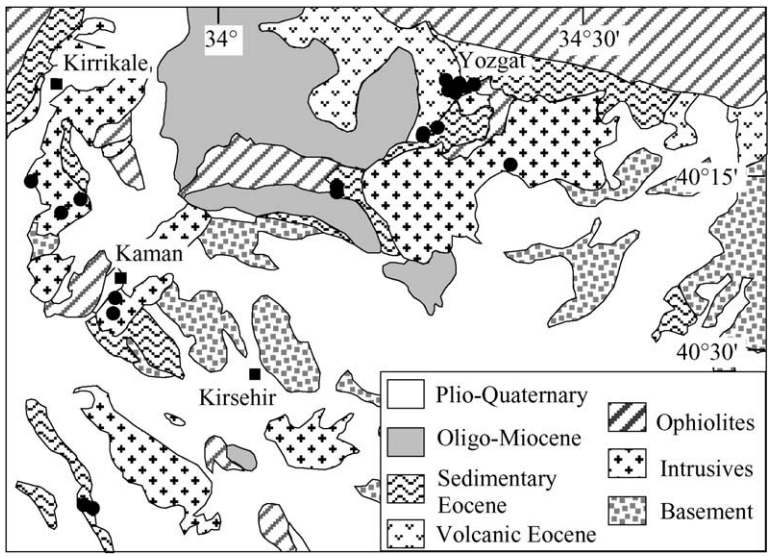


Fig. 3. Schematic map of the northern part of the Kirsehir massif showing the location of the sampled sites close to Kaman and Yozgat.

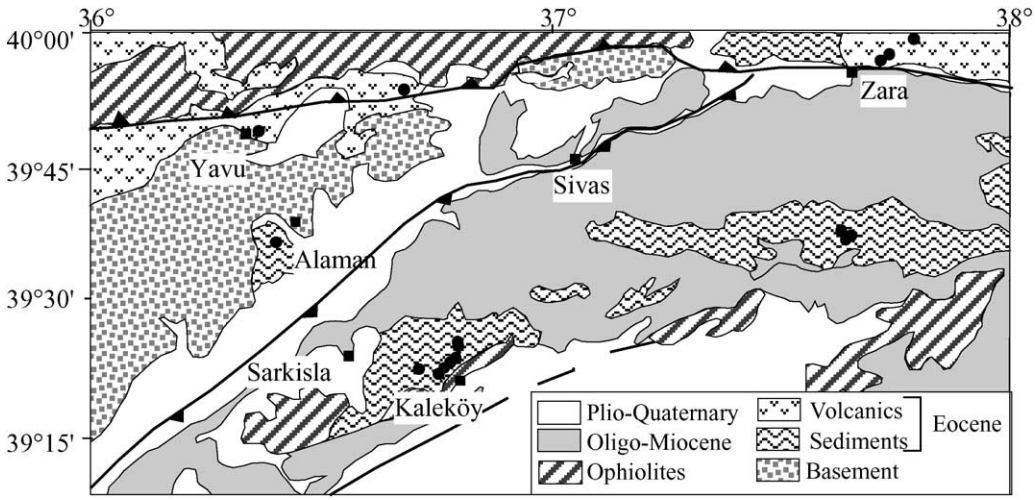


Fig. 4. Schematic map of the Sivas basin showing the location of the sampled sites.

(Fig. 1). This region is located in the rear of the thrust front of the ophiolitic nappes caused by the obduction of the south Neotethys oceanic crust. In the Antakya region, 38 sites were sampled in Eocene fine-grained limestones and in Miocene blue-grey marls. This region at the easternmost extremity of the Mediterranean Turkish coast is cut by the Amanos fault which can be considered as the northern termination of the left lateral Dead Sea fault, active since the middle–upper Miocene time. The sites are distributed on both sides of the fault. Three of the Miocene sites were sampled on the northern side of the Hatay mountain (Fig. 5).

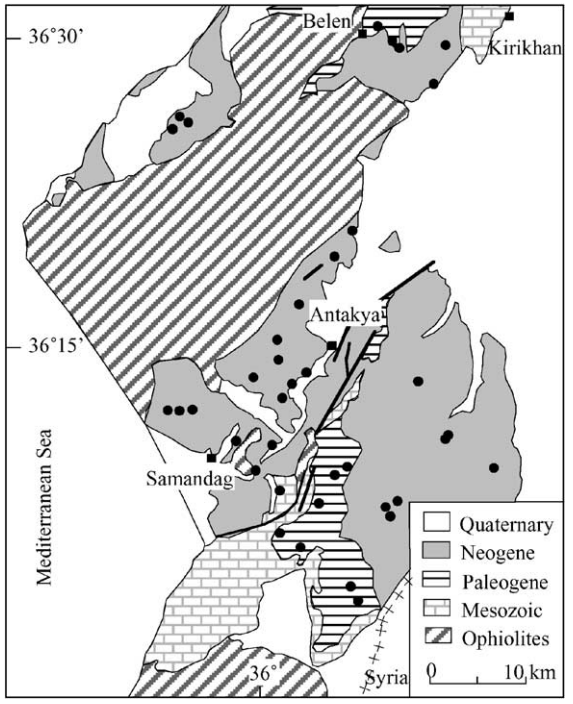


Fig. 5. Schematic map of the Antakya showing the location of the sampled sites.

4. Laboratory analyses

Stepwise acquisition of isothermal remanent magnetization (IRM) and subsequent AF demagnetization of the saturation IRM (SIRM) were systematically studied in all samples. In a large majority of the cases, 90% of the IRM was acquired in fields lower than 0.3 T, indicating that the main magnetic carrier is a low-coercivity mineral. Representative examples are given in Fig. 6a. In a few cases, a mixture of low- and high-coercivity minerals was observed (see, e.g. sample TK234 in Fig. 6a).

The anisotropy of magnetic susceptibility of the sedimentary samples has been systematically analyzed. Most of the samples are characterized by an oblate magnetic fabric with a minimum axis of anisotropy, K_{min}, perpendicular to the bedding plane, indi-

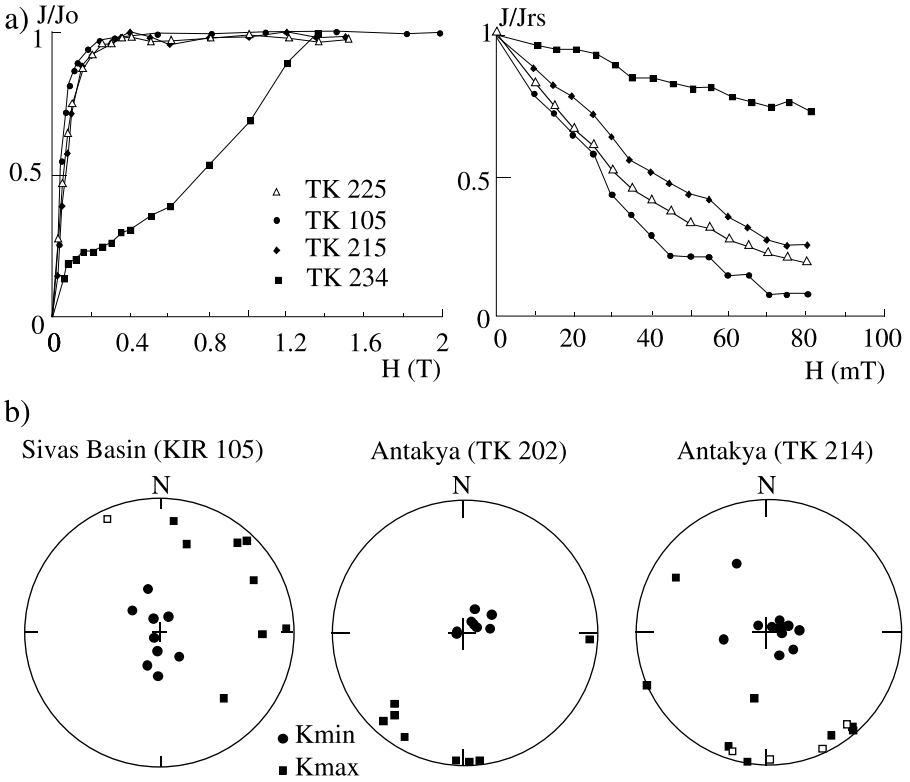


Fig. 6. (a) Typical SIRM acquisition and demagnetization curves. Most of the samples saturate at field of about 0.2–0.3 T. A few examples illustrated here by sample TK 234 showed a mixture of magnetic minerals. (b) Representative stereographic projections of the maximum (K_{max}) and of the minimum (K_{min}) axes of anisotropy obtained from the Turkish sedimentary sites.

cating deposition in low-energy environments (Fig. 6b).

Unfortunately, the extremely low magnetization ($<10^{-5}$ A/m) of a significant fraction of the samples often hindered reliable measurements even using a cryogenic magnetometer. In the Hatay region, for instance, 162 samples out of the sampled 414 (from 14 out of 38 sites) had to be discarded for this reason.

Stepwise thermal demagnetization has been systematically performed with 10–15 steps from room temperature up to a temperature depending on the sample (usually 500–550 °C). The low field susceptibility measured after each step of the thermal demagnetization usually did not undergo significant changes. Some samples, however, proved unstable during the thermal treatment (see, e.g. sample TK 119-04C in Fig. 7) which was stopped after the onset of this erratic behavior. In addition, at some sites, it proved impossible to remove a large overprint by the present-day field. These sites were, therefore, excluded from subsequent analysis.

5. Paleomagnetic results

The characteristic remanent magnetization (ChRM) was precisely determined by principal component analysis for 342 samples out of the 1300 studied samples (39 sites out of 113 sites) (Fig. 7). At each site, the mean direction was calculated using Fisher's (1953) statistic (Fig. 8) and reported in Table 1. The mean directions obtained from each region, reported in Table 2, are briefly discussed below.

5.1. The Pontides

Nine out of the sampled sites (18) yielded stable paleomagnetic directions. The three mean normal

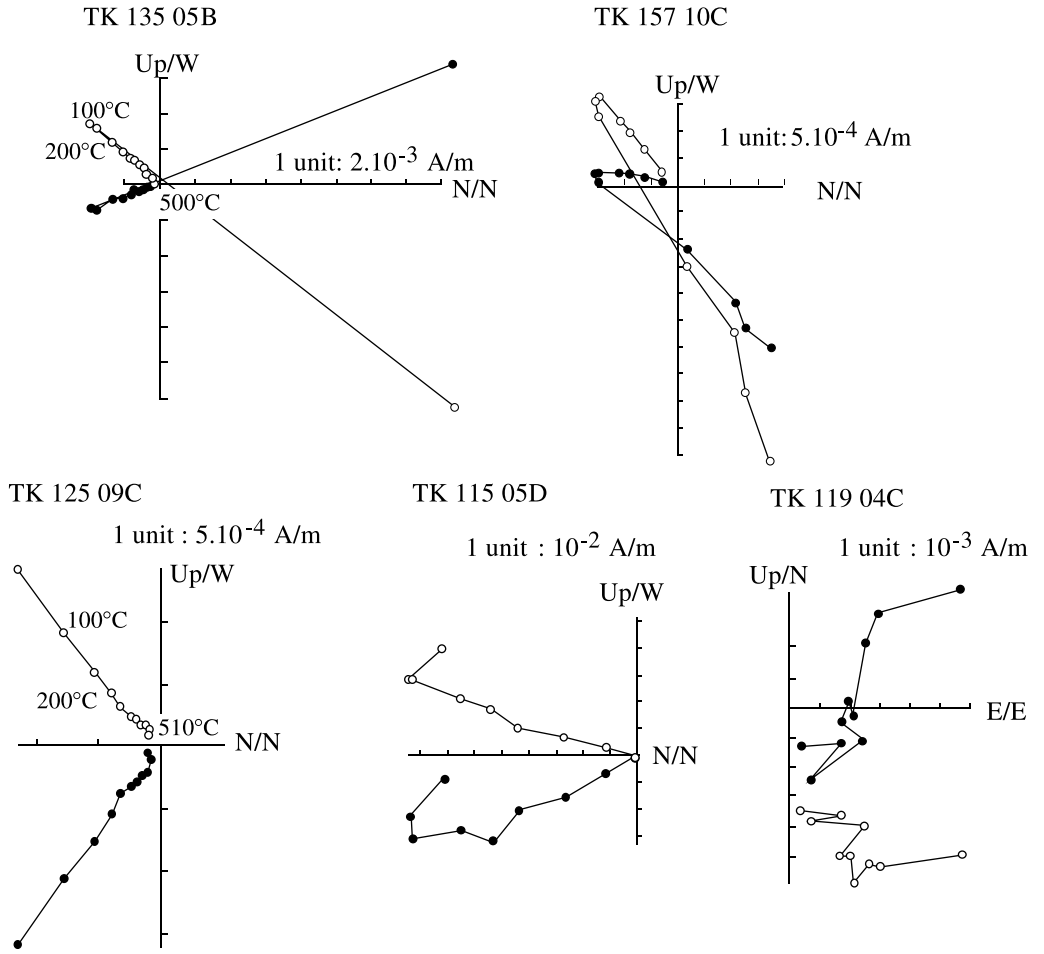


Fig. 7. Orthogonal projection of typical examples of demagnetization behavior. The white (black) dots are the projections onto the vertical (horizontal) plane. The diagram TK 119-04 C is typical for the noninterpretable diagrams.

directions are well opposite to the six reverse directions. After tilt correction (obtained from interbedded sediments for the volcanics), the results are quite consistent over the entire region. Declinations are westerly deviated by about 15° since Eocene time (Tables 1 and 2 and Fig. 9). Inclinations range between 26° and 45°.

5.2. The Kirsehir block

Three out of the five sites sampled in the Paleocene intrusives close to Kaman yielded stable directions with small within-site scatter. Declinations are all westerly deviated by ~45°, inclinations are clustered between 45° and 50° (Tables 1 and 2 and Fig. 10). Although no tectonic bedding attitude could be determined for these large plutons, this large deviation from the present-day field direction excludes a recent remagnetization.

Among the 17 sites sampled along the northern margin of the Kirsehir block, seven Eocene volcanic and sedimentary sites sampled close to the town of Yozgat yielded a well-defined mean direction (Tables 1 and 2 and Fig. 10). The Eocene volcanic flows and the interbedded sediments are located along the southern flank of a syncline and are characterized by a shallow northward dipping tectonic bedding accurately measured in the field. This tilt, however,

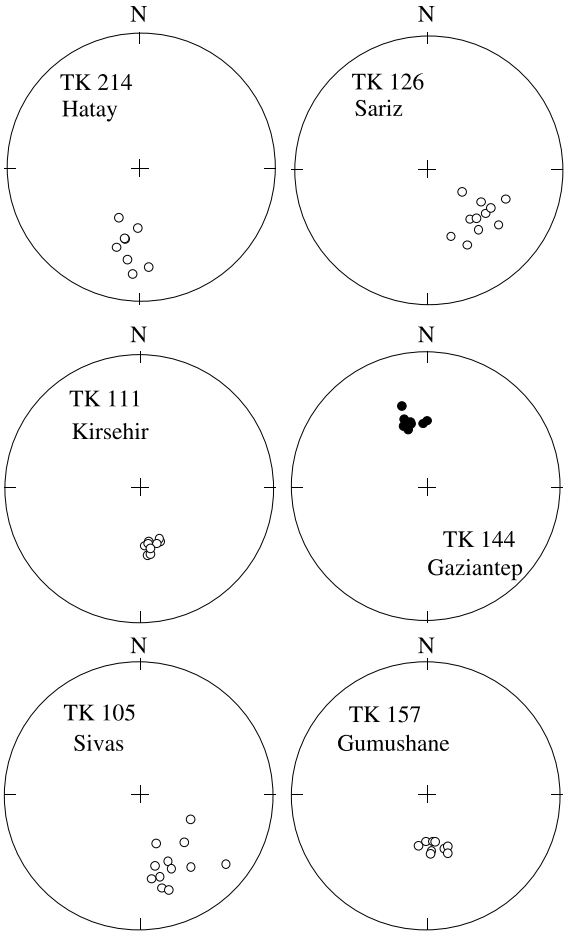


Fig. 8. Typical stereographic projection of the paleomagnetic directions obtained from single sites. White (black) symbols are for upward (downward) inclinations.

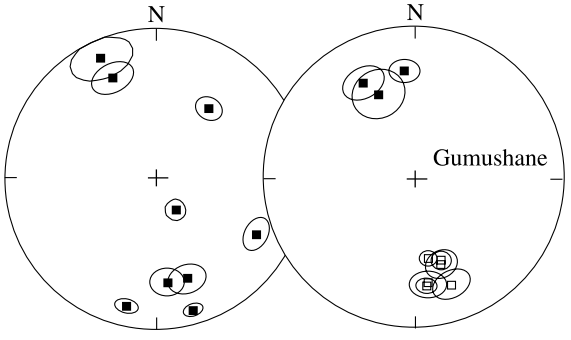


Fig. 9. Stereographic projection of the mean within site directions obtained from the Gumushane area with their confidence angle. Left-hand side: before bedding correction, right-hand side: after tilt correction.

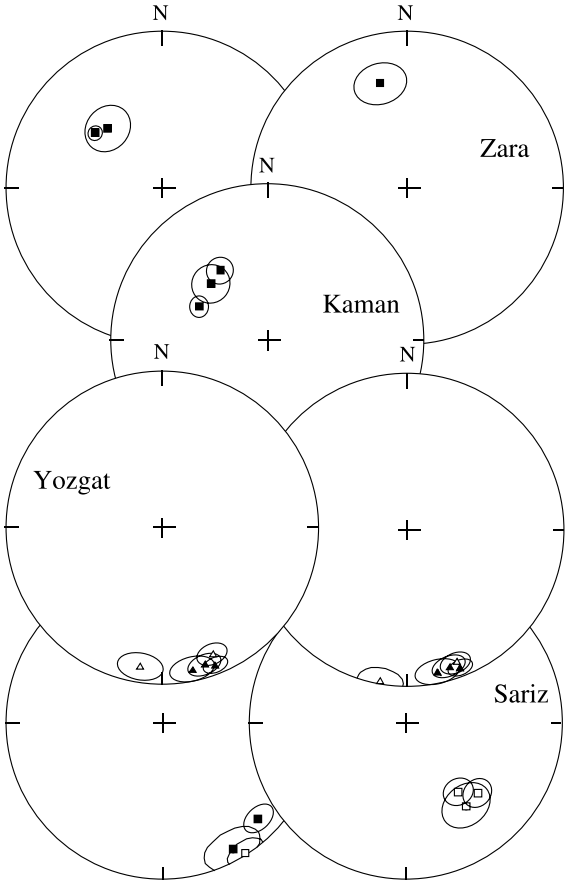


Fig. 10. Stereographic projection of the mean within site directions with their confidence angle obtained from the Sivas basin (Zara), the Kirsehir block (Kaman and Yozgat) and from the Tauric platform (Sariz). Left-hand side: before bedding correction, right-hand side: after tilt correction. The results from Kaman are reported in only one stereoplot because no tilt correction was applied.

is not large enough to allow a significant tilt test to be performed. It must be noted, however, that both before and after the correction, the directions differ from the present day field. Declinations are consistently counterclockwise deviated by about 25°. The shallow inclination at these sites (12°) is sharply inconsistent with values observed at other coeval Turkish sites.

5.3. Tauric platform (Sariz area)

In this region, reliable result were obtained from only 3 out of the 10 sampled sites. Fortunately,

however, the samples from these three sites are characterized by a perfectly stable behavior during thermal demagnetization, with small within-site scatter ($\alpha_{95} \sim 7°$) (Fig. 8). Directions pass the tilt test with the precision parameter k changing from 49 to 153 upon bedding correction (Tables 1 and 2 and Fig. 10). The mean direction calculated after tectonic correction has an inclination of 38° and a declination westerly deviated by 40° (Table 1).

5.4. Sivas basin

Two volcanic sites from the Zara region yielded a well-defined mean direction (Table 1 and Fig. 10). The paleohorizontal attitude of the lava flows has been obtained from interbedded sediments. Declination at these two Eocene sites are counterclockwise rotated by about 20° and the mean inclination is ~30°. On the other hand, sites in the Sarkisla area are all characterized by very unstable behavior during demagnetization and no reliable results were obtained.

5.5. Arabian plate

5.5.1. Gaziantep region

The characteristic remanent magnetization could be determined for only about one-third of the samples (from 7 out of the 13 sites). Two Miocene volcanic sites show well-grouped normal and reverse directions ($\alpha_{95} \sim 4°$), characterized by similar inclinations and westerly declination values (Table 1 and Fig. 11). The bedding attitude of these flows could unfortunately not be evaluated from field observations. However, a regional scale field investigation shows that none of the formations in this region has undergone tilting larger than 10°. Therefore, we used in situ paleomagnetic directions.

The directions obtained from upper Cretaceous–Paleocene calcareous and marly sites show both

Table 1
Reliable results obtained along a N–S transect in Eastern Turkey

Site	n/N	Age/ lithology	D_b	I_b	D_a	I_a	α_{95}	k
Gumushane region								
TK 148	10/10	E/v	174.0	31.0	173.0	−30.0	8.3	31
TK 150	6/10	E/l	193.0	13.6	174.0	−29.0	5.0	136
TK 151	7/11	E/v	336.6	28.0	332.0	29.0	10.0	28
TK 153	10/10	E/l	36.6	42.0	355.0	29.0	6.8	42
TK 156	8/11	E/v	335.6	13.0	337.0	39.0	13.7	13
TK 157	11/12	E/l	164.6	10.9	170.0	−45.0	4.0	109
TK 160	8/11	E/l	163.5	31.0	161.0	−26.0	9.0	31
TK 162	12/12	E/v	119.5	26.0	163.0	−40.0	8.0	26
TK 163	10/10	E/v	148.8	70.0	162.0	−42.0	5.3	70
Kirsehir region								
Yozgat								
TK 107	9/10	E/v	159.0	6.0	159.0	6.0	5.0	91
TK 108	10/14	E/v	163.0	9.0	163.0	9.0	6.8	42
TK 109	10/16	E/v	171.0	−47.0	149.0	−47.0	3.2	189
TK 112	11/14	E/l	168.0	7.4	168.0	7.4	9.0	22
TK 114	9/9	E/v	189.0	−10.2	190.0	−2.0	9.0	25
TK 115	9/12	E/v	159.0	−14.0	160.0	−10.0	6.5	51
TK 117	11/11	E/v	169.0	−42.0	178.0	−53.0	3.0	189
Kaman								
TK 118	7/9	P/i	315.0	48.0	315.0	48.0	10.0	26
TK 121	8/8	P/i	296.0	50.0	296.0	50.0	5.5	99
TK 122	5/10	P/i	326.2	45.3	326.2	45.3	7.0	75
Sariz region								
TK 126	12/18	E/l	135.0	14.0	134	−35.8	7.3	31
TK 128	5/6	E/l	151.0	8.5	144	−34.5	12.0	26
TK 129	10/16	E/l	147.5	−2.0	143	−43.5	7.5	35
Sivas basin (Zara)								
TK 100	10/10	E/v	309.0	43.8	337.0	32.0	3.7	139
TK 104	8/13	E/v	317.0	47.0	345.0	32.0	12.0	10
Gaziantep region								
TK 144	10/12	M/v	345.0	38.7	345.0	38.7	4.0	124
TK 145	10/13	M/v	131.0	−30.0	131.0	−30.0	4.4	99
Antakya region								
TK 215	8/8	E/l	352.5	49.3	9.5	54.0	4.5	108
TK 233	6/10	E/l	337.2	47.0	342.0	45.0	9.6	35
TK 201	12/13	M/m	167.7	−52.4	194.0	−52.0	7.0	32
TK 203	13/13	M/m	339.3	33.7	337.0	40.0	10.0	15
TK 208	9/10	M/m	352.0	56.0	3.2	46.5	5.3	76
TK 209	9/14	M/m	343.7	51.0	337.0	40.5	10.0	22
TK 211	6/15	M/m	339.5	37.0	344.5	39.5	11.0	27
TK 213	3/10	M/m	335.0	28.0	332.5	41.0	15.0	29
TK 214	9/12	M/m	185.0	−30.0	189.5	−41.0	7.5	37
TK 220	4/9	M/m	349.5	42.7	340.5	41.7	9.5	54
TK 225	8/12	M/m	182.5	−42.0	182.0	−40.0	10.3	23
TK 227	7/13	M/m	317.0	32.0	318.0	41.7	9.3	32
TK 232	10/11	M/m	165.5	−50.0	165.0	−33.0	6.0	60

Notes to Table 1:
n/N: number of reliable directions taken into account in the calculation/total number of samples analysed in the same site. Age: P: Paleocene, E: Eocene, M: Miocene. Lithology: l = limestones, i = intrusives, v = volcanics, m: marls. D_a, I_a, D_b, I_b: declination and inclination values before and after tectonic correction is applied. α_{95} and k: statistical parameters (Fisher, 1953).

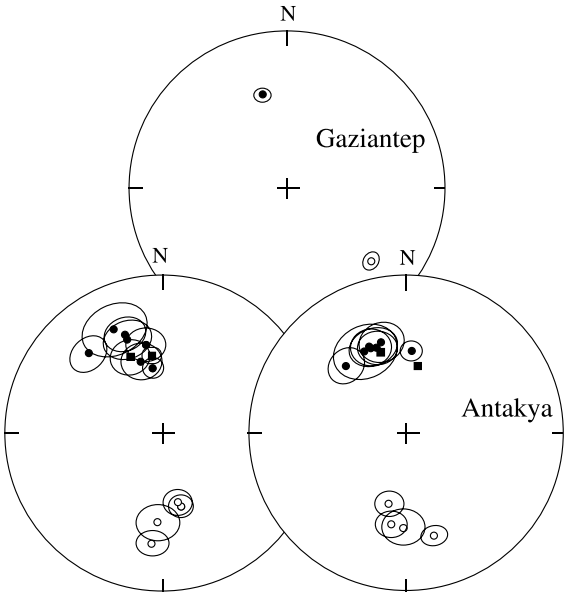


Fig. 11. Stereographic projection of the mean within site directions obtained from the northern boundary of the Arabian platform with their confidence angle. No tilt correction was applied for the Gaziantep sites, while the directions before (left) and after (right) tilt correction are reported for the Antakya region.

reverse and normal polarities. The two groups are antiparallel before bedding correction. However, scatter increases upon bedding correction (k decreases from 96 to 11) indicating that the magnetization postdates the formation of the syncline, i.e. it was acquired after early Miocene time. Therefore, no precise age can be given to the magnetization. Unfortunately, no other paleomagnetic data with better age control are available from this area.

5.5.2. Antakya area

Only 13 out of the 38 sedimentary sites sampled in this area yielded interpretable results (Table 1 and Fig. 11). Nine of these sites have normal directions and four have reverse directions with some scatter in the data. The average regional directions, both before and after tectonic correction, are slightly different from the present-day field. As tilts are of the order of 15°, the tilt test was not, however, really significant. The mean direction after tectonic correction has a westerly deviated declination ($D = 348 \pm 7°$) with an inclination of about 44°.

6. Discussion

This study aimed to reconstructing the Cenozoic geodynamical evolution of Anatolia constitutes the only paleomagnetic survey made on such a large geographic scale. Despite the large coherence observed at the different sites, the relatively small number of sites yielding reliable results makes it, however, clearly hazardous to base such a reconstruction only on the results reported here. Fortunately, a coherent picture of the Cenozoic evolution of the region appears when these results are combined with those already reported by other authors. In a general way, each of these previous studies concern an area more restricted than that of our study, but the areas studied by the different authors are distributed over Anatolia. The mean directions from the different regions are reported in Table 2.

6.1. Paleolatitudes

To interpret the data in terms of rotation and/or latitudinal displacement of Anatolia with respect to African and European plates, we have used the Cenozoic apparent polar wander paths (APWP) proposed by Besse and Courtillot (1991) for these two plates. In Fig. 12 and in Table 2, we report for every studied region the difference ($\Delta\lambda$) between the Eocene to Quaternary paleolatitudes inferred from the inclination data and the present latitude of the sites. In

Notes to Table 2:

Ages: Up. K. (upper Cretaceous), Pa (Paleocene), Eo (Eocene), Ol (Oligocene), Mio (Miocene), Plio (Pliocene), Q (Quaternary). N: number of sites taken into account in the statistical calculation. R and N are for reverse and normal polarity, respectively, in the sections studied for magnetostratigraphy by Krijgsman et al. (1996). D, I: declination and inclination. α_{95} and k: statistical parameters. $\Delta\lambda$: difference between the paleolatitudes deduced from the inclination data and the present latitude of the sites.

[a] Not taken into account in the calculation of the regional mean direction (see text).

Table 2
Cenozoic paleomagnetic results recalculated from site mean data given in the literature for Eastern Turkey

Region	Age	N	D	I	α_{95}	k	$\Delta\lambda$	Refs.
Pontides								
Central Pontides	Eo	3	146.4	−34	19.0	29	−23	Saribudak (1989)
Eastern Pontides	Up. K	7	166.8	−50	10.0	31	−10	Orbay and Bayburdi (1979)
	Up. K	6	153.5	−37	11.0	35	−20	Van der Voo (1968)
	Eo	6	346.0	40	7.0	90	−18	Van der Voo (1968)
	Eo	9	345.0	35	6.2	68	−22	This study
Kirsehir massif								
Kaman	Pa	10	148.0	−61	8.7	17	3	Sanver and Ponat (1980)
	Pa	3	313.0	48	16.0	62	−10	This study
Yozgat	Eo[a]	7	167.0	−13	22.3	8	−33	This study[a]
	Eo	5	158.4	−47	18.6	18	−12	Tatar et al. (1996)
South	mid-Mio	7	170.0	−58	10.9	27	0	Gürsoy et al. (1998)
	Plio	13	174.3	−52	8.6	23	−5	Gürsoy et al. (1998)
Karapinar	Q	5	156.9	−42	14.9	25	−13	Gürsoy et al. (1998)
Nevsehir–Akasaray	5–12 Ma	2	336.0	43	15.7	255	−13	Platzman et al. (1998)
	2–5 Ma	7	352.8	66	27.2	6	10	Platzman et al. (1998)
	0–2 Ma	6	356.0	57	9.4	52	0	Platzman et al. (1998)
Cappadocia	Q	6	189.7	−53	16.0	18	−4	Tatar et al. (2000)
Kayseri	Q	15	351.5	51	7.5	27	−7	Tatar et al. (2000)
NAFZ								
Tokat–Niksar	Up. K	2	140.0	−36	–	–	−20	Van der Voo (1968)
	Eo	11	150.0	−50	13.0	13	−10	Platzman et al. (1994)
	Eo	18	149.0	−45	6.8	27	−14	Tatar et al. (1995)
Erbaa	Eo	7	194.6	−49	15.3	16	−11	Piper et al. (1996)
Amasya	Eo	5	152.6	−44	17.9	19	−14	Piper et al. (1996)
Gerede	Pa	1	347.5	54	–	–	−6	Lauer (1981)
	7–21 Ma	14	344.0	59	11.6	13	−1	Platzman et al. (1994)
Sivas basin								
Zara	Eo	2	341.0	32	–	–	−23	This study
Yeniköy	Ol (29 Ma)	R	126.3	−33	8.8	6	−21	Krijgsman et al. (1996)
		N	329.6	48	7.2	13	−10	Krijgsman et al. (1996)
Inkonak	Ol (24 Ma)	R	129.8	−37	7.7	13	−18	Krijgsman et al. (1996)
		N	307.1	50	10.0	8	−9	Krijgsman et al. (1996)
Gemerek	Mio (15 Ma)	R	112.0	−61	9.6	7	3	Krijgsman et al. (1996)
		N	307.9	32	6.7	12	−21	Krijgsman et al. (1996)
Sarkisla	Eo	11	155.2	−53	9.9	15	−6	Gürsoy et al. (1997)
	Mio	14	145.5	−54	12.0	12	−5	Gürsoy et al. (1997)
	Plio	5	155.0	−52	7.8	29	−7	Gürsoy et al. (1997)
	Q	11	152.1	−49	17.4	11	−9	Gürsoy et al. (1997)
	13–18.1 Ma	3	332.7	35	31.2	17	−19	Platzman et al. (1998)
	4–5.8 Ma	3	351.4	55	16.0	61	−3	Platzman et al. (1998)
Tauric platform								
Sariz	Eo	3	140.0	−38	10.0	153	−17	This study
Tunceli	Eo	4	152.5	−48	16.0	33	−10	Van der Voo (1968)
Northern boundary of the Arabian plate								
Gaziantep	Mio	2	328.0	34	–	–	−18	This study
Antakya	Eo	2	345.0	48	–	–	−7	This study
	Mio	11	347.0	43	7.6	36	−11	This study

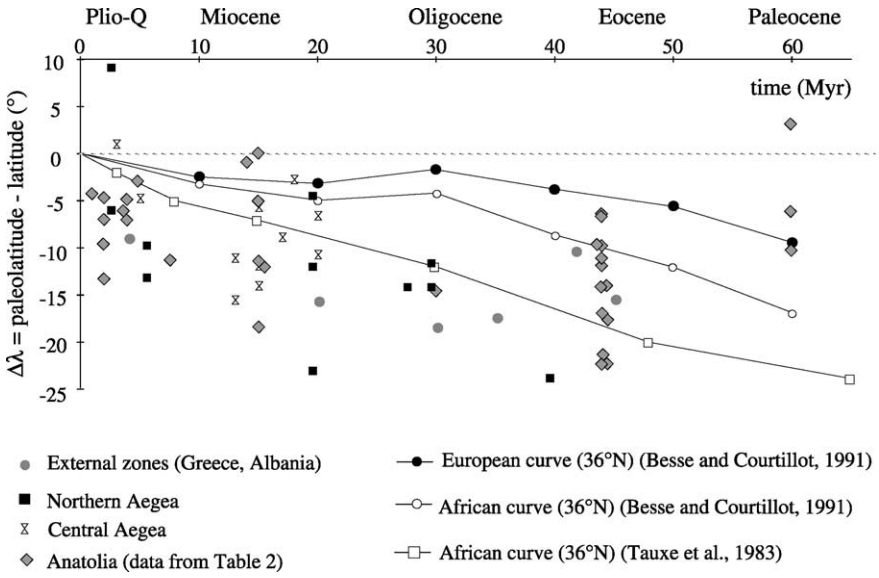


Fig. 12. Difference between the paleolatitude calculated from the paleomagnetic data and the present latitude of each studied region in the Aegean realm and in Turkey. Only the results obtained from igneous formations and from limestones which are reliable inclination recorders have been reported. The reference curves for Europe and Africa are calculated using the Apparent Polar Wander Path given by Tauxe et al. (1983) and Besse and Courtillot (1991) for a region located in central Turkey (36°N, 36°E).

Fig. 12, we compare these differences with the African and European reference curves calculated for a region located in central Turkey (latitude: 36°N; longitude: 36°E) (Besse and Courtillot, 1991). It clearly appears that the $\Delta\lambda$ values for Turkey are significantly different from both reference curves (Fig. 12). The shallow inclinations observed at the Turkish sites would suggest that the sampled formations have undergone a northward displacement of some ~15–20° since their formation. This large displacement is not, however, geologically realistic: it would indicate that Turkey was located at low latitudes, south of the present northern margin of Africa.

Similar shallow inclinations have been also reported from the Aegean realm (Kissel and Laj, 1988; Westphal, 1993), and their origin has been the subject of an active debate (Beck and Schermer, 1994), which unfortunately did not lead to a clear understanding of the problem. In sedimentary formations, the inclination error may be substantial for some lithologies (Laj et al., 1982), but in other lithologies and for volcanic formations, this kind of artifact cannot be invoked. Yet, shallow inclinations are documented also from volcanic formations. The hypothesis of a large-scale geomagnetic anomaly, although appealing, has been considered unconvincing by Beck and Schermer (1994) because such a strong anomaly should be seen in rocks of similar age worldwide. The hypothesis that these anomalous inclinations arise from an undetected general tilt in the Aegean–Turkish domain can confidently be rejected because the bedding attitude of most volcanic sites could be precisely determined by interbedded sediments, and the widely scattered nature of the sampling areas makes this possibility totally unrealistic. Finally, although the references curves used above are the most recent, they could still be in error. We have, therefore, also used the estimate for the African curve given by Tauxe et al. (1983), which had been used in connection with the Aegean results (Kissel and Laj, 1988). As shown in Fig. 12, the resulting inclination anomaly is reduced in this case. However, most of the sites still plot below the African curve. Although different scenarios have been suggested, we believe that the problem of shallow inclinations is not really understood. Despite this problem, the different authors have considered that declinations can be reliably used to infer possible rotations around

vertical axis. In the following, we shall also discuss the results along these lines.

6.2. Rotations around vertical axis—geographic pattern

It appears from Table 2 that the mean declinations obtained from the different regions and for the different investigated periods of time are coherently counterclockwise deviated with the only exception of a few sites located in the Erbaa area within the NAFZ and a few sites of recent age in Cappadocia. Following other authors, we will discuss the results in term of rotation using these declination values as a measure of rotation. It should be noted, however, that according to the APWP proposed by Besse and Courtillot (1991) for the European and the African plates, about 8–10° should be added to the Tertiary declinations to describe the movement of Anatolia with respect to the major plates (Fig. 13).

In the eastern Pontides, our results indicate counterclockwise rotations of some $15 \pm 7.5°$ since the Eocene time. These results are consistent with those of the first paleomagnetic survey dedicated to Turkish tectonics conducted by Van der Voo (1968), as early as 1968, who obtained the paleomagnetic direction from six Eocene sites ($D = 346°$, $I = 40°$, $\alpha_{95} = 7°$) and from six upper Cretaceous sites ($D = 153°$, $I = -36°$, $\alpha_{95} = 11°$) (Table 2). A similar upper Cretaceous direction had been obtained from dykes by Orbay and Bayburdi (1979) (Table 2). All these results, therefore, consistently document counterclockwise rotations of about 15° since Eocene. Saribudak (1989), who partially used for eastern Pontides some of our unpublished data, noted that similar directions are observed from the upper Cretaceous sites from the

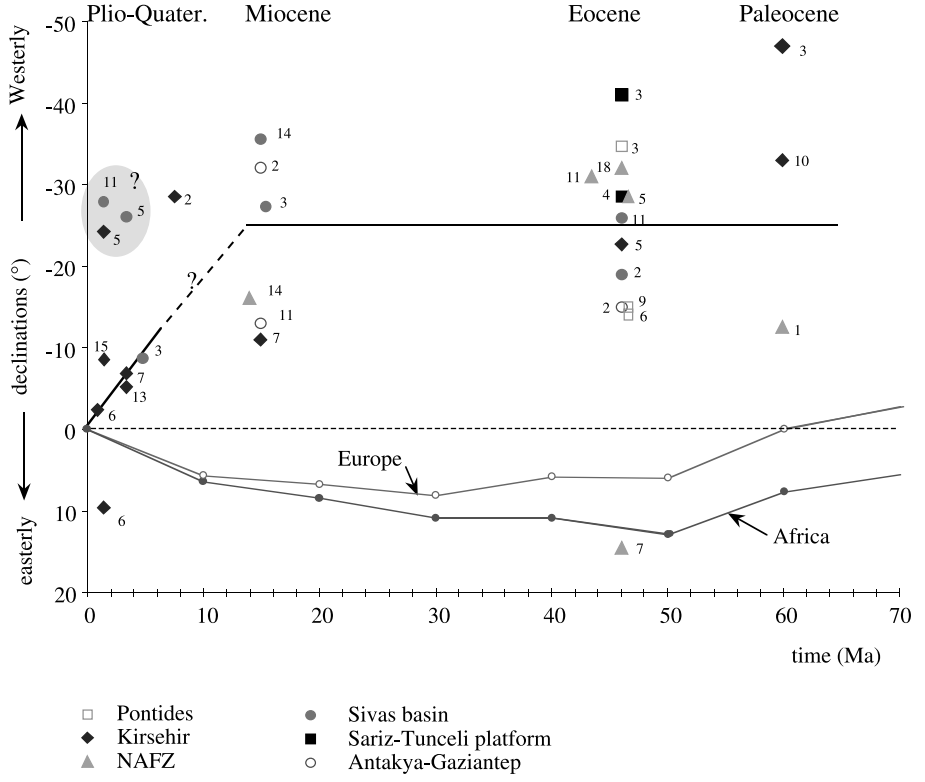


Fig. 13. Declination angles obtained from the different Anatolian geological/tectonic units versus time (the results from Krijgsman et al., 1996 are not taken into account for the reason given in the text). The reference curves for Europe and Africa are calculated using the Apparent Polar Wander Path given by Besse and Courtillot (1991) for a region located in central Turkey (36°N, 36°E).

western Pontides and Eocene sites from the Central Pontides, while in the central part between Zonguldak and Samsun, no significant rotation is measured (Table 2).

We did not obtain results from the area between the Pontides and the Kirsehir massif, but Platzman et al. (1994) obtained a post-Eocene counterclockwise rotation of about 30° very consistent with the results obtained previously by Van der Voo (1968) from upper Cretaceous formations (Table 2). Similar result were also obtained in the same area by Tatar et al. (1995) and Piper et al. (1996). However, while Platzman et al. (1994) did not observe significant rotations specifically related to the motion along the fault, Piper et al. (1996) found easterly declinations from the Erbaa area located within the fault zone. It must be noted, however, that both the declinations and the inclinations reported in this last study are largely scattered and that, in addition, the precise location of the sites is not reported in the original paper. Although the 14 Miocene directions from the Gerede region in the central part of the NAFZ are also scattered, the regional average directions also suggest counterclockwise rotations on both sides of the NAFZ (Platzman et al., 1994) consistent with the result obtained from a single site by Lauer (1981) in the same area (Table 2). Therefore, although small-scale block rotations most probably occurred in the immediate vicinity of the fault, no significant difference is observed north and south of the main fault zone. The main rotation does not appear to be distributed across the fault zone.

In the Kirsehir massif, our results from the Kaman Paleocene intrusive rocks yield a characteristic direction defined by $N=3$, $D=313°$, $I=48°$, $\alpha_{95}=16°$. The declination is consistent with the results of Sanver and Ponat (1980) from the same area ($N=10$, $D=328°$, $I=61°$, $\alpha_{95}=8.7°$). However, the mean inclination determined by these authors is larger than the one we obtain from the same area and also from the one obtained from other early Tertiary sites from Turkey (Table 2). In a recent study from Eocene units in the Yozgat area, Tatar et al. (1996) obtained values for the inclinations that are more consistent with the present day latitude than the shallow inclinations we report here. This discrepancy between the two sets of data is not understood. The results of Tatar et al. (1996), although characterized by significant scatter ($\alpha_{95}=18°$), indicate a post-Eocene counterclockwise rotation of about 23°. The same angle of rotation is obtained by Platzman et al. (1998) from Miocene formations while the Pliocene and Quaternary sites studied by these authors indicate on the average a small angle of rotation of about 8–4°. The Quaternary declinations reported by Gürsoy et al. (1998) and Tatar et al. (2000) in the same area are less informative because they are largely scattered, ranging between 337° (Gürsoy et al., 1998) and 10° (Tatar et al., 2000) (Table 2).

In the Sivas basin, our study yields a paleomagnetic direction of $D=341°$, $I=32°$ from Eocene volcanic rocks. This direction is obtained from only two sites but is consistent with the direction obtained by Gürsoy et al. (1997) for the same period and from the same region. In the same area, three Oligocene and Miocene sections were analysed for magnetostratigraphic studies by Krijgsman et al. (1996). The sections are well dated but the paleomagnetic results are of difficult to use for tectonic purposes because in each section the normal and reverse directions are far from being antipodal (Table 2). We prefer, therefore, not to use these results which were, however, considered by Platzman et al. (1998). Gürsoy et al. (1997) also analyzed Miocene, Pliocene and Quaternary rocks from this region. They observed that declinations have similar values ($D \sim 150°$) for all these three periods (Table 2) and interpreted this (see below) as an indication of a very recent rotation of some 30° arising from brittle crustal block rotations between major thrusts and strike-slip faults following in time a crustal thickening. These results, however, are significantly inconsistent with those obtained by Platzman et al. (1998) from much better dated formations. These last results indicate that in this area middle Miocene declinations are westerly deviated by about 27° while Pliocene declinations are only westerly seeking by about 5°. Consequently, Platzman et al. (1998) concluded that the main rotational phase occurred in the late Miocene period.

In the Tauric platform, our results from Sariz indicate a declination westerly deviated by about 40°, slightly larger than the value previously reported by Van der Voo (1968) from four Eocene sites located at Tunceli, south of the town of Erzincan (Table 2).

Along the northern margin of the Arabian platform, our results document westerly deviated declinations.

In the Antakya region, similar values of 345° are obtained from Miocene and Eocene sites. These are the only paleomagnetic data reported from this area so far.

6.3. Rotations around vertical axis—time evolution

Although data from the different regions are rather scattered, a coherent picture of the Cenozoic evolution of Anatolia appears when the results are plotted versus time (Fig. 13).

Starting from the oldest sampled sites, the average Paleocene declinations were obtained from only three groups of data (from a total of 14 sites) and they are distributed between 347° and 313°. Given this very limited number of data, the average counterclockwise rotation documented cannot be precisely evaluated.

For the Eocene period, which is the most widely investigated period of time in Turkey, average declinations range between 320° and 345°. A scatter of some 20° is generally observed within any particular sampled geologic unit. This scatter and the imprecise age of the sampled unit do not allow to detect a possible evolution of the angle of rotation.

During middle–late Miocene, the data are also largely scattered but one may be tempted to identify two groups of sites, one with a large rotation of about 30°, the second with a more limited rotation of about 10–15°. However, the number of sites and the large scatter makes this distinction in two groups speculative. The only quantitative result is the average rotation calculated from all sites which is closely the same as that from the Eocene sites.

Finally, results from the most recent (Pliocene–Quaternary) formations clearly identify two groups of rotation: in one group which comprises 16 Pliocene and Quaternary sites from the Sivas basin (Gürsoy et al., 1997) and 5 Quaternary sites from Karapinar volcanics in the southern part of the Kirsehir massif (Gürsoy et al., 1998), rotations are of the same order as for the Miocene/Eocene sites. In the other group, 44 sites studied by Platzman et al. (1998), Tatar et al. (2000) and Gürsoy et al. (1998) from the Kirsehir massif and the Sivas basin, document much smaller rotations (5–10°).

Based on the first group, Gürsoy et al. (1997) suggested that the counterclockwise rotation of Eastern Anatolia may have occurred more recently than previously believed. However, the sites characterized by this large recent rotation are located in the same regions as those belonging to the second group which document a much smaller amplitude. For example, the mean Pliocene declination of 335° obtained by Gürsoy et al. (1998) from the Sivas basin is much larger than the Pliocene declinations observed nearby (~ 351°) (Platzman et al., 1998). A possible explanation for this inconsistency may be found in the presence in these regions of numerous strike-slip faults. Therefore, one cannot rule out the possibility that the largest rotations result from very local block rotation linked to the activity of these faults, similar to the sites situated in the NAFZ.

The close similarity in the amplitude of the rotation of Eocene and Miocene rotations indicate that these rotation occurred during the Neogene, although it is not precisely bracketed in time.

This large consistency of the Eocene and Miocene results all along the N–S Turkish transverse, from the eastern Pontides to the Arabian platform also clearly suggests that the different blocks welded into a single coherent unit at least by the Miocene, in agreement with the conclusion of Platzman et al. (1994, 1998). Following Platzman et al. (1998), we believe that Pliocene results indicate that the rotation started after the Miocene. We do not, however, following these last authors, identify three possible phases of rotation at different periods. The scatter of the data appears to be too large to discriminate between these phases. In particular, the second phase of rotation defined by Platzman et al. (1998) during upper Miocene (5–12 Ma) is based only on two points yielding large rotations (>50°). One of them is from Krijgsman et al. (1996) and we decided not to use it here, as discussed above, because of the severe lack of antipodality between normal and reverse directions, the other is from Platzman et al. (1998) and it is clearly out of the main trend defined by older and younger sites obtained by these authors in the same area.

The results discussed above show that the welded blocks have rotated by an angle of about 25° in a counterclockwise sense during the Neogene. Assuming that this happened in a semi-rigid mode around the pole defined by MacClusky et al. (2000) (31°N, 32°E) on the basis of the geodetic data, this rotation corresponds to a large displacement which reaches about 500 km at an average latitude of 40°N. Assum-

ing further that it is associated with the Arabian collision and started about 12 Ma ago, its amplitude corresponds to an average displacement of about 40 mm/year at 40°N. For the Kirsehir massif, the Sivas basin and the Tauric platform located between the two major discontinuities of the NAFZ and the EAFZ, this displacement is slightly larger but consistent with the 28 mm/year instantaneous displacement deduced from the GPS survey by Le Pichon et al. (1995) and the 20–24 mm/year given by MacClusky et al. (2000). This may suggest that the average movement observed over the last 12 Ma is still active and slightly slower today.

The most unexpected conclusion which can be drawn from the paleomagnetic results is that not only the regions located between the NAFZ and the EAFZ have rotated but also the Pontides and the northern boundary of the Arabian plate. As observed by Platzman et al. (1994), the rotation on both sides of the NAFZ is the same, implying that the North Anatolian Fault itself may have rotated. This point already suggested by Van der Voo (1968) is confirmed by the results reported here. The geographical extension of the region which has undergone the rotation, therefore, suggests that the NAFZ constitutes a very narrow zone of brittle deformation accommodating only part of the rotation documented by the paleomagnetic data. The main decoupling area between the mobile Turkish unit and Europe, precisely placed for the most recent period along the NAFZ appears to be situated further to the north and, but cannot be precisely located from our results for the finite rotation over a few million years.

7. Conclusion

The Neogene counterclockwise rotation of Anatolia is one component of the pattern of rotation in Eastern Mediterranean. It can be combined with the results obtained previously from the Aegean realm (Laj et al., 1982; Kissel and Laj, 1988; Speranza et al., 1995; Kissel et al., 1995). On this broader geographical scale, the paleomagnetic results obtained from Eastern Turkey to the Adriatic and Ionian Seas indicates that the main trend of the geodynamic evolution of this area can be described in terms of two rotations of opposite sense. A clockwise rotation of the western part of the

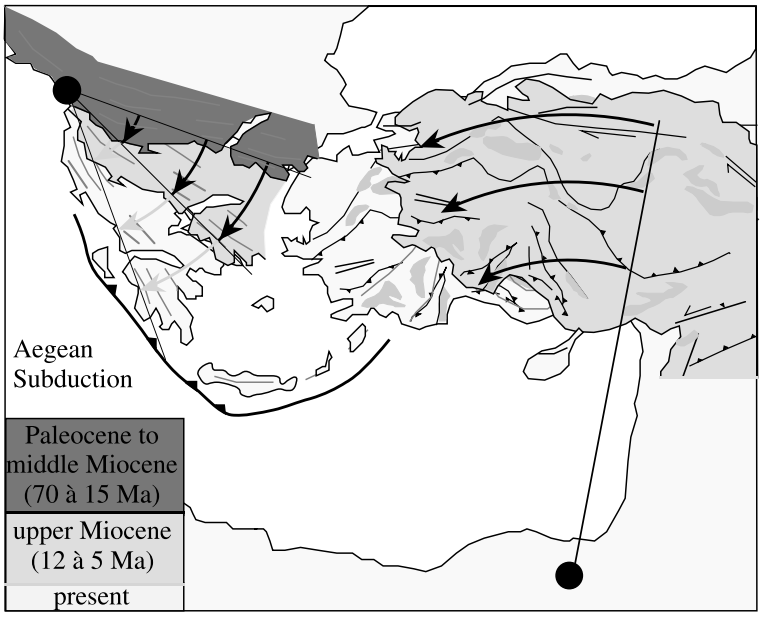


Fig. 14. Schematic evolution of the eastern Mediterranean deduced from the paleomagnetic data and showing the two rotations in opposite senses in western Greece and in Turkey.

Aegean arc around a pole located at the vicinity of the Scutari-Peç transverse zone occurred in two phases: first a middle Miocene rotation of some 25° (Kissel and Laj, 1988; Speranza et al., 1995), then a second rotation of roughly the same amplitude during Plio-Pleistocene (Laj et al., 1982; Kissel and Laj, 1988; Duermeijer et al., 1999; Kissel, 2001; Duermeijer and Langereis, 2001). The Dalmatian zone in Montenegro and Croatia was a fixed area for the investigated period of time (Kissel et al., 1995).

The counterclockwise rotation observed in eastern Turkey started when the first clockwise rotation has stopped in western Greece. Accompanying the Arabian collision, the transfer of the westward displacement associated to the rotation of Anatolia may have plaid a role, inducing the youngest phase of rotation at the western side of the Aegean realm. These two phases of rotation are schematically illustrated in Fig. 14.

It is worth noting that according to the GPS data obtained over the last 12 years, the rotations are still active today on both sides of the Aegean realm. The main difference between the Neogene rotations determined by paleomagnetic studies and the GPS results is that the regions located north of the main discontinuities (Pontides, north of the North Anatolian Fault in Turkey and Epirus, Corfu, Albania, north of the Kephallinia Fault Zone in Greece) appear to be fixed today while they have undergone large amplitude finite rotation.

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